

MAYANK MISHRA

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PROFESSIONAL EXPERIENCE

Doctoral Candidate

Multispeech, LORIA

December 2024 – Present 📍 Nancy, France

About: The PhD is about semantic segmentation of sound scenes and events for edge devices. Worked on Task 4 of the [DCASE challenge](#).

NLP masters Intern

INRIA

Apr 2024 – Aug 2024 📍 Nancy, France

About: The focus of the internship was to benchmark several sound separation deep neural network (DNN)-based algorithms according to their carbon energy footprint, to find a robust and efficient model for the Universal sound separation task. This model can be further used in combination with speech enhancement systems to aid automatic speech recognition (ASR).

Machine Learning Intern

LIMICS

Jun 2023 – Aug 2023 📍 Paris, France

About: Internship on understanding the CoAT algorithm for analogical transfers and metric learning. Approach towards getting theoretical and experimental results on improving the efficiency of the algorithm.

Senior Content Developer

Cerebry

May 2021 – Sep 2022 📍 Kanpur, India

Core responsibilities: Analysis, structuring and development of content for the online AI-based platform of the company.

Additional responsibilities: Task allocation, supervision and troubleshooting for a team of 7 members and training of new employees in the content group.

Skills Utilized: Python programming, mathematics, team work, effective communications and leadership.

SKILLS

- **Programming Languages:** Python, R, C++, HTML
- **Query Languages:** SQL, SparkQL
- **Software Tools:** LaTeX, Mathematica, Matlab, MongoDB, Git
- **Packages:** Pandas, Numpy, Sci-kit Learn, Keras, PyTorch, Tensorflow
- **Natural Languages:** Hindi (Native), English (Bilingual Proficiency), French (A2)

EDUCATION

Master in Natural Language Processing | IDMC, Université de Lorraine, France

Sep 2022 – Sep 2024 📍 Nancy, France

Bachelor of Science-Master of Science, Dual Degree | IISER Bhopal | Mathematics (Major) & Physics (Minor)

2013 – 2018 📍 Bhopal, India

PROJECTS

AI for Indian Farmers - Handwritten Devanagari

Fruitpunch AI

Oct 2024 – Jan 2025 📍 Nancy, France

- Challenge aimed at creating an accurate OCR solution for images of handwritten Devanagari script.
- Implemented GAN-based image enhancement for low-quality input data to the OCR model.

AI for greener data-centers

Fruitpunch AI

Jun 2024 – Aug 2024 📍 Nancy, France

- Challenge aimed at using reinforcement learning to reduce power consumption of cooling systems in data centers.
- Utilises various data analysis techniques and models for reinforcement learning. Worked towards improving the evaluation metric for RL systems used in the task.

Blogs' Posts

Data Mining

Sep 2023- Jan 2023 📍 Nancy, France

- Project aims at using fundamental concept analysis (FCA) to answer questions about blog posts' data.
- Utilises various data cleaning, analysis and mining techniques used in Natural Language Processing.

Music genre recognition

Artificial Neural Networks

Nov 2023- Jan 2023 📍 Nancy, France

- The project task includes classification of musical genre based on input audio spectrograms (treated as images)
- Utilized Convolutional Neural networks.

PUBLICATIONS

- [Metric Analysis for Spatial Semantic Segmentation of Sound Scenes](#) (Pre-print), Mayank Mishra, Paul Magron and Romain Serizel
 - [Description and Discussion on DCASE 2025 Challenge Task 4: Spatial Semantic Segmentation of Sound Scenes](#), Masahiro Yasuda, Binh Thien Nguyen, Noboru Harada, Romain Serizel, Mayank Mishra, Marc Delcroix, Shoko Araki, Daiki Takeuchi, Daisuke Niizumi, Yasunori Ohishi, Tomohiro Nakatani, Takao Kawamura, Nobutaka Ono
 - [The lambda number of the power graph of a finite p-group](#), Mayank Mishra and Siddhartha Sarkar
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